NNDL project report template

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***Abstract*—I suggest to write the Abstract as the very last thing. You may sketch it at the beginning, but then always finalize it at the end.**

***Index Terms*—Neural Networks, Convolutional Neural Networks, Generative Adversarial Networks, Symbolic-Domain Music**

I. Introduction

The task of generating music is a frontier in Artificial Intelligence, since it poses the challenge of training a network to be creative, but with the rigid constraints imposed by music theory. While some architectures are already capable of generating technically correct sequences, capturing the key points of realistic, human-like compsitions, remains an interesting topic. This is a praticularly interesting problem if seen in the scenario of creating assistive tools for composers: both experienced and non experienced ones would benefit by these tools in order to improve creativity and exploration of genres. What we propose is an architecture that aims no only to model the placement of notes, but also the evolution of the themaic they represent.

In music generation, coherence is the main objective. That's why a simple RNN architecture seems to be the most tappropriate solution, as it's structure aims at learning sequences in order to predict the next values. However, MidiNet [1] showed that convolutional GANs can overcome the issue. The main problem is that in this particular configuration, the discriminator tends to be better than the generator, eventually quickly reaching a stable 100% accuracy. The network that we propose tries to enhance the capabilities of MidiNet by adding to the generation process a set of residual layers unit that should help in learning long-term relations. Additionally, as suggested in MidiNet paper, we changed the loss function from a conventional cross-entropy to a Wasserstein loss, and in particular we chose the version that uses gradient penalty [2], [3]. The idea is to avoid vanishing gradients, which easily arise from the previous loss function. This changes aim to improve the quality of the game played by Generator and Discriminator, forcing more balanced capabilities to both the structures, which should eventually lead to a more realistic result.

Summary of contributions:

- WGAN-GP (Wasserstein GAN - Gradient Penalty): loss is modified to help longer and more complex relations to be learned
- Residual Units: The generator is enriched with residual units to provide more direct flow of the gradient.
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In the paper we follow this structure. In Section II we describe the base model we started from, in section III we explain the dataset we used and the preprocessing techniques we adopted. In Section IV we explain our model and we conclude in Section V showing the results we obtained.

II. Related Work

III. Processing Pipeline

IV. Learning Framework

V. Results

VI. Concluding Remarks

References

[1] L.-C. Yang, S.-Y. Chou, and Y.-H. Yang, "MidiNet: A Convolutional Generative Adversarial Network for Symbolic-Domain Music Generation," in *Proceedings of the 18th International Society for Music Information Retrieval Conference (ISMIR)*, (Suzhou, China), Oct. 2017.

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[3] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein GAN," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, (Sydney, Australia), Aug. 2017.

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